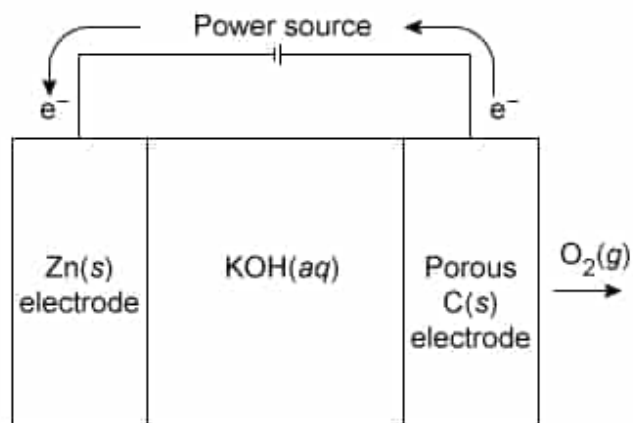


A charging Zn-O₂ battery is shown below.



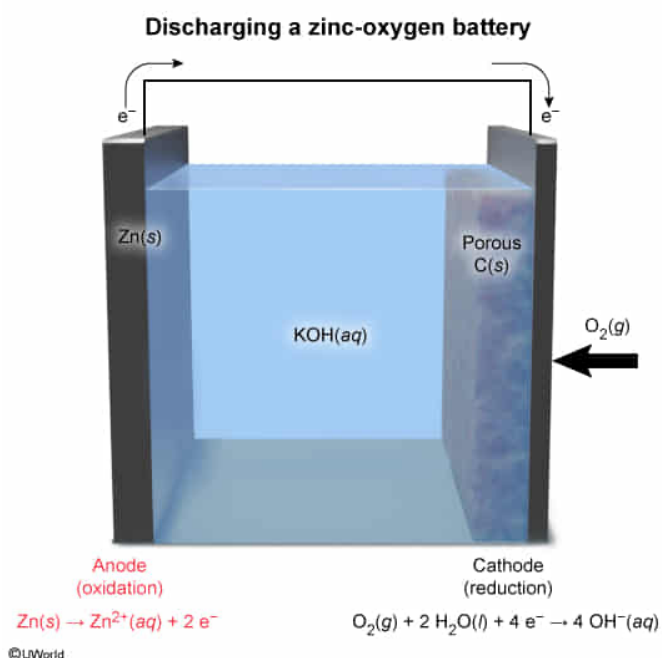
Which of the following reactions occurs at the anode as the battery is *discharging*?

- ☐ A. $\text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4 e^- \rightarrow 4 \text{OH}^-(aq)$ [8%]
- ☐ B. $4 \text{OH}^-(aq) \rightarrow \text{O}_2(g) + 2 \text{H}_2\text{O}(l) + 4 e^-$ [29%]
- ✓ ☒ C. $\text{Zn}(s) \rightarrow \text{Zn}^{2+}(aq) + 2 e^-$ [47%]
- ☐ D. $\text{Zn}^{2+}(aq) + 2 e^- \rightarrow \text{Zn}(s)$ [14%]

Correct

46% answered correctly

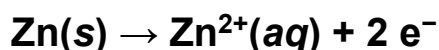
Explanation:



When a battery is **charging**, an electric current supplied from an external source is used to drive a **nonspontaneous** reduction-oxidation (**redox**) reaction (ie, the battery operates as an **electrolytic cell**). Conversely, when a battery is **discharging**, a **spontaneous** redox reaction occurs, producing an **electric current** (ie, the battery operates as a **galvanic cell**).

Although electron flow is in the **opposite direction** when a battery is charging versus when it is discharging, electrons always flow from the **anode**, where **oxidation** takes place, to the **cathode**, where **reduction** takes place. In effect, the electrodes switch roles when the cell reverses from charging to discharging.

According to the diagram shown in the question, as the Zn-O₂ battery is charging, electrons flow from the C(s) anode to the Zn(s) cathode. Consequently, as the battery is discharging, the electrons flow from the Zn(s) anode to the C(s) cathode (ie, the opposite direction). Because oxidation happens at the anode, Zn(s) is oxidized as the battery is discharging:



(Choice A) As the battery is discharging, the C(s) electrode acts as the *cathode*, where oxygen is reduced.

(Choice B) Oxygen is oxidized at the C(s) anode as the battery is *charging*.

(Choice D) In this reaction, Zn(s) is *reduced*. *Oxidation* occurs at the *anode*, not reduction.

Educational objective:

When a battery is discharging, it behaves as a galvanic cell; when it is charging, it behaves as an electrolytic cell. In either case, oxidation occurs at the anode and reduction occurs at the cathode, and electrons flow from the anode to the cathode.

Time spent: 0 sec

Subject:
General Chemistry

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System:
Solutions and Electrochemistry